Weather Impacts on Electricity Demand and Grid Operations

Heat, cold, wind, precipitation, humidity, and severe-weather considerations

Document purpose: This document is prepared as a searchable knowledge-base source for a SmartGrid AI RAG prototype. It is an educational synthesis, not an official ERCOT, utility, or government publication.

Overview

Weather affects both how much electricity customers use and how reliably parts of the electric system can operate. The direction and size of the effect depend on season, location, building characteristics, customer behavior, generation mix, and infrastructure condition.

Extreme weather can create a compound challenge: it may increase demand at the same time that it reduces the availability of generation, fuel, or transmission equipment. A forecasting system should therefore consider weather variables together rather than treating temperature as the only driver.

Extreme heat

During hot periods, residential and commercial air-conditioning use commonly increases. Higher temperatures can raise peak electricity demand, particularly when hot conditions persist into the evening and buildings remain warm. Humidity can increase discomfort and cooling needs, while cloud cover and solar output patterns can also affect the generation-demand balance.

Heat can also affect equipment performance and operating limits. High temperatures may reduce the efficiency or capability of some equipment and can increase thermal stress on power lines and transformers. The exact impact depends on equipment design, loading, maintenance, and local conditions.

Extreme cold

Cold weather can increase electricity use through space heating, especially in buildings that rely on electric heating or heat pumps. Sudden cold snaps may cause demand to rise quickly. In some regions, cold weather can also affect fuel supply, power plant operations, and equipment that is not adequately weatherized.

Cold-weather risk is not limited to demand. Ice, freezing precipitation, and very low temperatures can affect infrastructure and access for repair crews. A reliable analysis should distinguish cold-related demand effects from cold-related supply and network risks.

Wind, storms, and precipitation

Strong winds, thunderstorms, ice, and heavy precipitation can damage distribution lines, poles, substations, and vegetation near electrical equipment. Storms can also make repairs slower or more difficult. Precipitation itself does not automatically imply an outage; location, intensity, duration, wind, lightning, flooding, and asset exposure all matter.

Wind has a dual role in power systems. It can contribute to electricity generation where wind farms are connected, but severe or highly variable wind conditions can also create forecasting and operating challenges. A model should not assume that higher wind speed always increases or always decreases grid stress.

Feature engineering guidance

Useful weather features may include daily or hourly mean, maximum, and minimum temperature; humidity; wind speed; precipitation; snowfall; and lagged values. Derived features can include cooling-degree measures, heating-degree measures, rolling averages, and indicators for extreme conditions.

Weather features should be aligned with the prediction horizon. For example, a next-day demand model should use weather forecasts that would have been available at prediction time, not future observed weather. This prevents data leakage and makes evaluation more realistic.

Key takeaways

- Heat and cold can increase electricity demand through cooling and heating.
- Severe weather can affect demand, generation, transmission, and distribution at the same time.
- Wind and precipitation effects are context-dependent; avoid simplistic assumptions.
- Use weather forecasts available at prediction time to prevent data leakage.

References and further reading

• NOAA National Weather Service — https://www.weather.gov/

• NERC Reliability Assessments — https://www.nerc.com/our-work/assessments

• ERCOT Grid Information — https://www.ercot.com/gridinfo

Prepared 9 November 2026. Verify current operational information with the linked official sources before making real-world decisions.